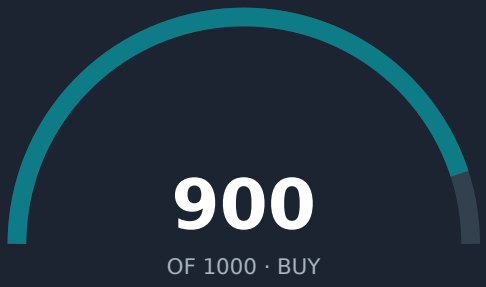
BUILD SPECIFICATION + CLAUDE MASTER PROMPT

SCORE1000

AI market-scoring app for iPhone & iPad



10-stage analysis · model-per-stage plan · trade plans with 3–5 take-profits
server-side monitoring · emergency protocol · iCloud sync · Liquid Glass UI

Prepared for Devendra Jondhale · 4 November 2026 · Working name — rename freely

What this document contains

#	Section	What you get
1	Read this first — where your plan needs correcting	Ten honest corrections before any code is written
2	How the app works end-to-end	Flow, architecture diagram, why a server is mandatory
3	Which AI model at which stage	Fast / Balanced / Deep tiers, stage-by-stage mapping, cost per run
4	Scoring method & trade plan	How 10 stage scores become 0–1000, verdict bands, TP ladder
5	Data sources & every market you can trade	Free vs paid APIs, India + global, product universe
6	Apple-specific: Liquid Glass, Stocks app, background limits	What iOS allows and what it doesn't
7	Security, SEBI & App Store rules	Real-money safeguards and legal boundaries
8	Your setup checklist & roadmap	Accounts, keys, milestones, paper-trading gate
9	The Claude master build prompt	Copy-paste prompt (also delivered as a separate .md file)
10	Sources	Where the facts in this document come from

How to use it

Read sections 1–8 once (about 20 minutes). Then open Claude Code in an empty private GitHub repo, copy **Score1000_Claude_Build_Prompt.md** into the chat, add this PDF to **/docs/**, and build one milestone at a time.

1. Read this first — where your plan needs correcting

You asked me to be straight with you. The idea is strong and buildable, but ten parts of the brief would either not work as written or would put real money at risk. Each item below says what to change and why. The rest of the document already uses the corrected version.

#	What you wrote / assumed	What's actually true and the fix
1	**A 900/1000 score is not a 90 % chance of profit**	Recent benchmarks of top AI models predicting stock direction found roughly 60 % accuracy over 3 days, falling to about 51 % — a coin flip — at 10 days (PriceSeer, 2026). The score is a structured checklist, not a forecast. **Fix:** the app records the outcome of every analysis and shows, per score band, how often it actually worked ("calibration"). Trust the score only after it has a track record on your products.
2	**AI must not read candlesticks directly**	Language models make 10–20 % errors on multi-step calculations even in good conditions. Feeding thousands of raw candles also costs a lot. **Fix:** code computes every indicator, pattern, level and statistic; the AI receives a compact feature summary and explains it. Every number shown is code-computed.
3	**"Entire history at 5-minute bars" is impossible and unnecessary**	Free Indian minute data starts from January 2022 (Upstox), and decades of 5-minute bars would be millions of tokens per run. **Fix:** full history at monthly/weekly/daily, recent windows for intraday (e.g. 60 days of 5-min), all reduced to statistics in code.
4	**Equal weighting of the 10 stages is wrong**	Leadership quality matters for a 2-year hold and almost nothing for a 5-minute scalp. **Fix:** weights depend on the holding horizon (table in section 4). You can still pick "Equal" to see your original method.
5	**Buy/sell is not decided only in step 10**	If stages score without a direction, step 10 has nothing consistent to add up. **Fix:** every stage scores on one scale: 1 = strongly bearish, 50 = neutral, 100 = strongly bullish. The composite gives Buy score and Sell score together, plus a "No trade" zone in the middle — the most important verdict of all.
6	**An iPhone cannot monitor prices every minute**	iOS gives no reliable 1-minute background execution, and Apple rejects tricks to keep apps alive. **Fix:** a small server (yours, in India, about the cost of a mobile recharge per month) watches the market and pushes alerts to the phone. This is the biggest change to your plan.
7	**An emergency notification is not a stop-loss**	Push notifications can be delayed, Focus modes can hide them, and you may be asleep. **Fix:** the app tells you to place a real stop-loss / GTT order at your broker when you enter; alerts are the second line of defence.
8	**"Free live data for all markets" doesn't exist**	Upstox and Fyers give free Indian data with an account. Zerodha's free tier has **no** market data. Global real-time data is free only within tight limits (Twelve Data: 800 calls/day). NSE forbids redistributing its data. **Fix:** a provider-per-market routing table, caching, and a clear paid-upgrade path (~$30–60/month) if needed.
9	**Scanning "top products" with all 10 AI stages will burn money**	Running the full pipeline on hundreds of symbols costs real dollars per scan. **Fix:** AI Recommendation mode uses a cheap code screener first, a fast model to shortlist, and the full 10-stage pipeline only on the best 5–10.
10	**This must stay a personal app**	Giving buy/sell calls to others for any consideration requires SEBI Research Analyst registration, and Apple expects trading/investing apps on the App Store to come from licensed companies, not individuals. Auto-placing orders brings in SEBI's 2025 retail-algo rules (static IP, Algo ID, daily 2FA). **Fix:** v1 is analysis-only, for you, via internal TestFlight; you place trades yourself in your broker app.

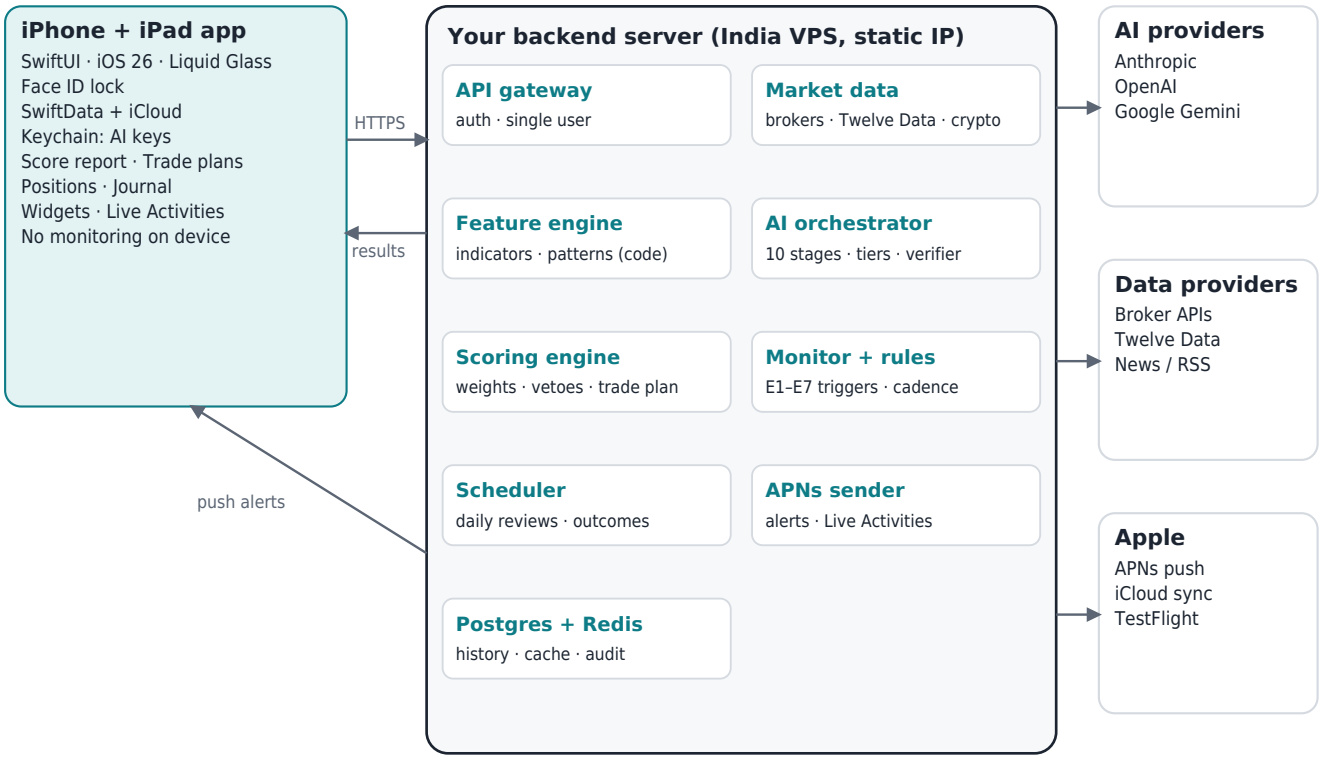
Two ideas I'm adding that you didn't ask for

Paper-trade mode by default. Run the app on paper for 4–8 weeks, let it build its calibration record, then switch to real money with Face ID. This is the cheapest insurance you can buy.

A verifier pass. After the final AI verdict, a second, cheaper model (ideally from a different company) re-checks every number and level against the computed data. Disagreement downgrades the verdict to "No trade".

2. How the app works end-to-end

The app has two halves. The **iPhone/iPad app** is where you search, read reports, confirm trades and get alerts. The **backend server** does the heavy and always-on work: market data, feature computation, AI calls, monitoring and push notifications. Both are yours — no third party sees your trades.



Flow of one analysis

1. You pick a product (e.g. RELIANCE on NSE, XAUUSD, AAPL) and a timeframe/holding style. The timeframe sets the weights and the monitoring cadence.
2. The server pulls candles across 1M → 5m, news, events and related products, and builds a **feature pack** in code.
3. Ten stages run (some in parallel). Each returns a 1–100 score, a confidence, reasons that cite evidence, and sources.
4. The scoring engine combines them into a 10–1000 composite, applies vetoes, and stage 10 + code produce the trade plan.
5. A verifier re-checks the numbers. You see the full report: score gauge, verdict, each stage's "why", and the exact maths.
6. You can save it to **Upcoming Trades**. The server watches the entry zone and alerts you when price comes near.
7. When you mark it as entered (price, quantity), it becomes a **Position**. The server monitors it at the right cadence, sends TP-hit and emergency alerts, and re-scores it daily/weekly with continue / reduce / exit advice.
8. When you close it, the journal records invested amount, P&L, R-multiple and whether the AI was right — feeding calibration.

Monitoring cadence follows your timeframe (as you asked)

Timeframe chosen	What the server does	AI use	How you're told
1 min / 5 min (scalping)	Live broker WebSocket; rules on every tick and bar close	None per tick — rules only. Fast model writes the alert text after a trigger	Live Activity on Lock Screen + time-sensitive push
15 min / 1 hour	Rules at every bar close	Fast model on triggers	Push
Daily	Price-alert levels live; full review after market close	Balanced model, batch (cheaper)	Push + morning digest
Weekly / Monthly	Price-alert levels live; daily close check; weekly deep re-score	Balanced daily (light), Deep weekly	Daily digest + emergency push

3. Which AI model at which stage

You're right that not every stage needs the most expensive model. The rule I used: **use the cheapest model that can do the job, and spend on depth only where reasoning quality changes the decision.** Classification and reading computed features is cheap work; explaining multi-year history and making the final call is expensive work.

The three tiers

Tier	Anthropic (Claude)	OpenAI	Google Gemini	Good at
Fast	Haiku tier — claude-haiku-5-5 $0.10 in / $0.50 out per 1M tokens	nano / mini tier (e.g. gpt-5-nano)	Flash-Lite tier	Classifying news, reading feature JSON, short summaries, verifier checks, alert wording
Balanced	Sonnet tier — Sonnet 5.5 $2 / $10	mid tier (e.g. gpt-5.4)	Flash tier	Event impact, sector news, relationships, 1-week outlook, daily position reviews, chat
Deep	Opus tier — claude-opus-5-5 $4 / $20	top tier (e.g. gpt-5.5)	Pro tier	Multi-year causal history, final multi-timeframe verdict and trade plan, weekly deep review

Anthropic prices are from Anthropic's official pricing page (Nov 2026). OpenAI and Google model names come from dated third-party trackers and change often — that's why the app **never hard-codes model names**: after you paste a key, it lists the provider's live models and you map each tier to one. Anthropic's Fable/Mythos tier ($10/$50) exists but is not worth it here: the extra cost buys little over Opus for this task.

Stage-by-stage assignment

#	Stage	Code does first	AI tier	Web search	Refresh	Why this tier
1	Product news	Fetch NSE RSS / news APIs, dedupe	Fast	Optional	1–4 h	Many articles, simple per-article sentiment + impact; code aggregates
2	Upcoming events	Results dates, dividends, board meetings, macro calendar	Balanced	Yes	Daily	Judging how an event moves price needs reasoning
3	Leadership / governance / country	Promoter holding, pledges, ratings	Balanced	Yes	7–30 days	Slow-changing; cached so cost is low
4	Full monthly history + why	Regimes, drawdowns, biggest moves with dates	**Deep**	Yes	30 days	Causal "why did it rise/fall" across years is the hardest reasoning; runs rarely
5	Weekly, 6 months	Patterns + this symbol's own hit-rates	Fast	No	Weekly	Interpreting computed features
6	Daily, full history	Indicators, backtested patterns	Fast → Balanced if signals conflict	No	Daily	Cheap by default, escalates only on conflict
7	Global sector news	Map product → sector/theme	Balanced	Yes	Daily, shared per sector	One run serves every stock in that sector
8	Connected / sister products	Rolling correlation, lead-lag	Balanced (map) + Fast (today)	Map only	Map 30 d, read daily	Knowing Reliance↔Jio or Gold↔USD is knowledge; today's reading is cheap
9	Today + 1-week outlook	VWAP, ranges, intraday structure	Balanced	Optional	Each run	Short-term synthesis of several signals
10	Final verdict + trade plan	Feature packs 5m→1M + stages 1–9	**Deep** + extended thinking	Yes (re-check)	Each run	The decision that touches money
V	Verifier	Compare AI levels with computed data	Fast (other provider)	No	Each run	Independent check catches hallucinated numbers

Other job	Tier	Note
Live monitoring (1m–15m)	None (code rules)	Never an LLM per tick — too slow, too costly, unpredictable
Daily review of each open position	Balanced via batch API	Batch is 50 % cheaper on Anthropic; runs after market close
Weekly deep re-score of open positions	Deep	Continue / reduce / exit with reasons
AI Recommendation mode	Code screen → Fast triage (top 50) → full pipeline (top 5–10)	Cost preview shown before it runs
"Explain this score" chat	Balanced	Uses the stored report as context

What one full analysis costs (estimate)

Based on an estimated ~104k input and ~14.5k output tokens per 10-stage run, at Anthropic's official prices. Replace with measured usage once the app logs real runs.

Model mix	AI tokens per symbol	Comment
All Fast (Haiku)	≈ $0.02	Cheapest, weakest reasoning on stages 4 and 10
Recommended mix (as in the table above)	**≈ $0.20–0.35**	Most of the quality of all-Deep at a fraction of the cost
All Balanced (Sonnet)	≈ $0.35	
All Deep (Opus)	≈ $0.71	Little extra benefit on simple stages
Web search add-on	+ $0.04–0.11	~8 searches per run; can cost more than the tokens on cheap models

Prompt caching (shared instructions and feature packs) and batch processing for daily reviews cut these further. Gemini prices are reported to rise in Dec 2026–Jan 2027, so the price table lives in an editable config file, and Settings has monthly and per-run spend caps.

4. Scoring method & trade plan

Your 10 × 100 = 1000 idea stays — with three refinements: a common direction scale, horizon-based weights, and confidence.

How the composite is calculated

- Each stage returns **score** 1–100 (1 = strongly bearish, 50 = neutral, 100 = strongly bullish) and **confidence** 0–1.
- Low confidence pulls a score toward neutral: `s_eff = 50 + confidence × (score − 50)`.
- Composite: `C = 10 × Σ(weight × s_eff) / 100`, giving 10–1000. Buy score = C; Sell score = 1010 − C.
- So "Buy 900" means strongly bullish across heavily weighted stages with high confidence; "Sell 900" is the mirror image.

Stage	Scalp / Intraday	Swing	Positional	Long-term	Equal (your original)
1 News	8	10	10	8	10
2 Upcoming events	4	8	10	10	10
3 Leadership / country	0	3	8	15	10
4 Monthly history	0	4	10	20	10
5 Weekly 6M	2	10	14	10	10
6 Daily	10	15	12	5	10
7 Global sector	4	8	10	12	10
8 Connected products	8	8	8	8	10
9 Today + 1 week	24	14	6	2	10
10 Final	40	20	12	10	10
Total	**100**	**100**	**100**	**100**	**100**

Starting weights, stored in a config file. Once you have outcome data, the app shows which stages actually predicted well so you can tune them.

Verdict bands and vetoes

Composite	≥ 750	620–749	381–619	251–380	≤ 250
Verdict	Strong Buy	Buy	No trade / Wait	Sell	Strong Sell

Any of these forces **No trade**, with the reason shown: stale data; results or a major event inside the holding window; poor liquidity or wide spread; verifier finds a mismatch; fewer than 7 of 10 stages

available.

Trade plan with 3–5 take-profit stages

Entry zone and stop-loss come from structure or ATR; R is the distance between them. Position size = (capital × risk %) ÷ R, default 1 % risk per trade. Default 5-stage ladder for a long trade:

Stage	Target	Book	Then
TP1	1 R	30 %	Move stop to breakeven — the rest of the trade is now risk-free
TP2	2 R	25 %	Trail stop to TP1
TP3	3 R	20 %	Trail stop to TP2
TP4	4 R	15 %	Trail stop to TP3
TP5	Runner	10 %	ATR trailing stop

For 3 stages: 40 / 35 / 25 % at 1R / 2R / runner. Code snaps targets to nearby resistance and rounds to tick size, and rejects any plan with reward-to-risk under 1.5 at TP2.

Holding badges and complete-by date

Badge	Scalp	Intraday	Short-term	Swing	Positional	Long-term
Typical hold	Minutes	Same day	2–10 trading days	2–6 weeks	1–6 months	6 months +

Each trade shows a **complete-by date**. When it passes, the app asks: exit, or run a fresh analysis to justify extending.

Emergency protocol

Code	Trigger	Code	Trigger
E1	Stop-loss / plan invalidation crossed	E5	Re-score flips to the opposite side (thesis broken)
E2	Gap open beyond stop, or > 2 ATR against you	E6	Data feed or server heartbeat lost — "Monitoring offline"
E3	Fast adverse move: > 1.5 ATR within a few bars	E7	Complete-by date passed
E4	News shock: news score drops ≥ 25, or fraud / raid / SEBI order / default / CEO exit / trading halt		

Action: time-sensitive push repeated every 2 minutes until you acknowledge (max 5), red Live Activity, full-screen "Close now" screen, and a journal entry. E6 matters: silence must never look like "all fine".

5. Data sources & every market you can trade

Recommended stack (free first, paid upgrade later)

Need	Free / cheap choice	Limits that matter	Paid upgrade
NSE, BSE, F&O;, MCX, currency derivatives	**Upstox API** (free with account)	Minute/hour history from Jan 2022; daily/weekly/monthly from Jan 2000; 50 req/s; daily login token	Zerodha Kite Connect ₹500/app/month (free Personal tier has no data); Dhan Data API ₹499 + tax/month
India fallback	Fyers API (free Standard tier)	Live feed capped at 100 symbols; 5/s, 5,000/day; 5 yr daily, 1 yr intraday	Fyers Prime
US stocks, global indices, forex (incl. USD/INR), gold/silver	**Twelve Data** Basic	800 calls/day, 8/min; no Indian exchanges	Twelve Data Grow (from $29/mo) or EODHD ($29.99/mo)
Crypto	Binance public API + CoinGecko Demo	CoinGecko: 10,000 calls/month, attribution	—
Company news & filings	**NSE official RSS** (announcements, board meetings, results, circulars) + Finnhub or Marketaux	BSE's official feed is paid (reported ~₹9 lakh/yr)	Paid news API if needed
Macro calendar	FRED, US BLS schedule	US-centric	Paid economic calendar
Full product list	**Upstox instrument master** (one gzipped JSON of every NSE/BSE/MCX/F&O; instrument), global symbol lists	Refreshed daily by the server	—

Prices and limits are from provider pages or dated secondary sources (Jul–Nov 2026); a few could not be re-verified and are noted in section 10. Yahoo Finance's unofficial feed broke in 2023 and 2025 and its terms bar commercial use — fine only as a last-resort private fallback.

Every market the app will let you choose from

You asked for "all the products available online". Listing tens of thousands of symbols in a PDF wouldn't help you — instead, the app downloads the full, current instrument lists every day so the picker always shows everything tradable. These are the categories it covers:

Region	Asset class	Examples	Source
India	Equities (NSE + BSE), SME, ETFs	RELIANCE, TCS, HDFCBANK, NIFTYBEES	Upstox / Fyers
India	Indices	NIFTY 50, BANK NIFTY, SENSEX, FINNIFTY, sector indices	Upstox / Fyers
India	Futures & options (NFO, BFO)	NIFTY weekly options, stock futures	Upstox / Fyers
India	Commodities (MCX)	Gold, Silver, Crude Oil, Natural Gas, Copper	Upstox / Fyers
India	Currency derivatives	USDINR, EURINR, GBPINR, JPYINR	Upstox / Fyers
India	Mutual funds (NAV only)	Any AMFI scheme	AMFI NAV file (to verify)
Global	US stocks & ETFs	AAPL, NVDA, MSFT, SPY, QQQ	Twelve Data
Global	World indices	S&P 500, Nasdaq 100, Dow, FTSE 100, DAX, Nikkei 225, Hang Seng	Twelve Data
Global	Forex	EURUSD, GBPUSD, USDJPY, USDINR, DXY	Twelve Data
Global	Metals & energy	XAUUSD, XAGUSD, WTI, Brent	Twelve Data
Global	Crypto	BTC, ETH, SOL and other major pairs	Binance / CoinGecko

6. Apple-specific: Liquid Glass, the Stocks app, background limits

Can Apple's Stocks app be connected?

Short answer: no

Apple's Stocks app has no developer API, no shared watchlist access, no Shortcuts/App Intents for its data and no way for another app to read or write it. Being an Apple-ecosystem app doesn't change that. SCORE1000 uses its own data feeds instead.

What you get instead, all native Apple features: **Home Screen, Lock Screen and StandBy widgets** (score and P&L), **Live Activities and Dynamic Island** for open intraday trades, **Siri / Shortcuts** ("What's my score for Reliance?"), and **Spotlight** search for your products.

Liquid Glass design (iOS 26)

Liquid Glass is Apple's iOS 26 material: translucent, refracting surfaces that float above content. Used well it looks premium; used everywhere it hurts readability — fatal for a numbers app. The rule for SCORE1000: **glass for navigation and controls, solid surfaces for data.**

- System tab bars, toolbars, navigation bars and sheets adopt Liquid Glass automatically when built with Xcode 26 — no extra code.
- Custom floating controls (score chip, timeframe picker, quick-action bar) use `.glassEffect()`, grouped in a `GlassEffectContainer` so they blend and morph; transitions use `glassEffectID(_:in:)`.
- Buttons use `.buttonStyle(.glass)` and `.glassProminent` for the primary action.
- Never behind candlestick charts, tables, P&L or score breakdowns.
- When *Reduce Transparency* or *Increase Contrast* is on, the app switches to solid surfaces automatically.

Themes (your "not pressuring the eye" requirement)

Theme	Character
Liquid Glass (default)	System look with glass navigation; follows light/dark mode
Calm Light	Off-white background, slate text, softened green/red
Calm Dark	Deep grey (#121417) instead of pure black, low-glare accents
OLED Night	True black for night use, dimmed colours
Paper	Warm, low-blue tones for long reading sessions
High Contrast	Maximum legibility

Every theme has a colour-blind-safe option (blue/orange instead of green/red), uses tabular digits so numbers don't jump, and keeps motion subtle.

What iOS allows in the background — and what it doesn't

- **No reliable once-a-minute background task.** Background refresh only promises not to run *before* a time; in tests it ran after 20 minutes or not for hours.
- **Live Activities** update for up to 8 hours (one NSE session fits), up to 5 at once — ideal for intraday positions, not for monthly holds.
- **Time-sensitive notifications** break through Focus and Notification Summary — the main emergency channel.
- **Critical alerts** (sound even on silent) need a special Apple entitlement, historically granted to health and safety apps; we'll request it but not depend on it.
- So: the server watches, the phone displays. This is the correct architecture, not a workaround.

iCloud sync between iPhone and iPad

Analyses, watchlists, trade plans, positions and journal sync through SwiftData + your private iCloud database. API keys sync separately through iCloud Keychain (end-to-end encrypted). Secrets never go into the iCloud database. Large raw data stays on your server; the app keeps summaries.

7. Security, SEBI & App Store rules

Real-money security checklist

Area	Control
App lock	Face ID / passcode on launch and after 30 s in background; privacy cover in the app switcher; Face ID again to view keys, switch to real-money mode or export
AI keys	Keychain with iCloud Keychain sync; never in iCloud database, logs or crash reports. If the server runs daily reviews it holds an encrypted copy; Settings shows which keys the server has and can wipe them
Broker access	Broker tokens live only on your server; you approve the daily broker login from the app. No order placement in v1
Server	Indian region VPS, static IP, HTTPS only, firewall, SSH keys only, automatic security updates, nightly encrypted backups; single-user sign-in (Sign in with Apple + allowlist)
Network	App Transport Security on; certificate pinning to your own server only
AI safety	News and web text treated as untrusted data (prompt-injection defence); outputs schema-validated; numbers recomputed in code; AI has no ability to act on accounts
Audit trail	Every run stores model IDs, prompt version, data timestamps, raw and validated outputs, cost
Accounts	Two-factor on Apple ID, GitHub, broker, server host and every AI console; spend caps on AI keys
Lock Screen	Option to hide P&L in widgets and Live Activities

Legal boundaries (India)

I'm flagging these as a fellow-professional heads-up, not as legal advice — you'll judge the legal side better than I can.

Situation	What applies
Personal use (you, your devices, internal TestFlight)	No SEBI Research Analyst registration — RA rules cover research services "for consideration" as a business (RA Third Amendment, Dec 2024). No App Store review for internal TestFlight.
App ever places orders through a broker API	SEBI retail-algo framework (circular of 4 Feb 2025, fully in force 1 Apr 2026): broker's official API only, OAuth + 2FA, static IP whitelisted, daily re-login, exchange Algo ID; registration above 10 orders/second. Some sources say algos must be hosted in India.
Sharing scores/calls with others for any consideration	RA registration, AI-use disclosure (Reg 19(vii)), evidence-backed calls (Reg 20(4)), website, record-keeping. Sharing with family is an untested grey area.
Publishing on the App Store	Guidelines 3.2.1(viii) and 5.1.1(ix): trading/investing apps should come from a licensed legal entity, not an individual. 1.1.6: "for entertainment only" disclaimers don't excuse misleading output.
Market data	NSE retains ownership and bars redistribution without agreement; it has sent cease-and-desist notices to apps. Personal display only.

Built-in disclaimers

Shown in Settings and on every report: personal analytical tool, not investment advice, not a SEBI-registered RA/IA; AI and algorithmic scores can be wrong or outdated; trading involves substantial risk of loss; alerts depend on data, network and Apple delivery and may be delayed; market data © exchanges and vendors; TradingView attribution for charts.

8. Your setup checklist & roadmap

Things only you can do (Claude can't create accounts for you)

Item	What to do
Apple Developer Program	Enrol (reported ₹8,700/year in 2024 — confirm current fee). Needed for TestFlight, iCloud, push notifications.
App Store Connect	Create the app record; add yourself as an internal tester; create an App Store Connect API key (.p8) for GitHub Actions; create an APNs key (.p8).
GitHub	Private repo for the app + a second private repo for signing certificates (fastlane match); turn on 2FA.
Upstox	Developer app (API key, secret, redirect URL pointing to your server). Optional: Fyers as backup.
Twelve Data	Free API key for global markets.
AI providers	API keys for Anthropic (primary) and optionally OpenAI / Google; set monthly spend limits in each console.
Server	Small VPS in Mumbai or Bangalore with a static IP, plus a domain name for HTTPS.

GitHub → TestFlight → your iPhone

- GitHub Actions on Apple's `macos-26` runner builds with Xcode 26 (required for uploads since 28 Apr 2026).
- fastlane signs the app (match) and uploads with your App Store Connect API key to the **internal** TestFlight group — no Beta App Review.
- You install from the TestFlight app on iPhone and iPad. Builds expire after 90 days, so a scheduled monthly workflow re-uploads automatically.
- Alternative: Xcode Cloud (25 free hours/month with membership) — simpler signing, fine for a solo project.

Roadmap (each milestone ends with tests you can check)

Milestone	Delivers	You can verify
M1 Foundation	Design system, 6 themes, Face ID lock, Keychain, live model listing, CI → TestFlight	App on iPhone + iPad; keys sync; lock works
M2 Backend + data	Server, Upstox login, instrument master, candles, global/crypto, search, charts	Find RELIANCE, XAUUSD, AAPL, BTCUSDT; charts 1m→1M
M3 Feature engine	All indicators, patterns with hit-rates, feature packs	Values match hand-checked examples
M4 AI + scoring	10 stages, verifier, weights, vetoes, Score Report	Every reason cites evidence; cost per run logged
M5 Trades	Trade plans, TP ladder, sizing, Upcoming Trades, Positions, Journal	P&L matches your manual calculation
M6 Monitoring	Rules engine, push, Live Activities, widgets, emergency protocol	Simulated feed fires each alert within 10 s
M7 AI Recommendations	Screener → shortlist → full pipeline, cost preview	Stays inside spend cap
M8 Calibration + hardening	Outcome tracking, calibration charts, failure drills	App degrades clearly; never shows a made-up score

The gate before real money

Paper-trade for at least 4–8 weeks after M8. Look at the calibration screen: if "Buy 700+" calls haven't beaten a simple benchmark on your products, the weights need work before money goes in. That discipline is what separates this from a fancy guessing machine.

9. The Claude master build prompt

This is the same text as the separate **Score1000_Claude_Build_Prompt.md** file — use the file for copy-paste, since copying from a PDF breaks formatting.

0. Role and mindset

You are a principal iOS engineer, backend engineer and quantitative analyst working together as one team. You are building **SCORE1000**, a private AI market-analysis app for one user (Devendra) on his own iPhone and iPad, distributed only through internal TestFlight.

Real money depends on this app. That changes the priorities:

- Correctness, security and honest uncertainty come before features and visual polish.
- When a requirement is unsafe, impossible or unverifiable, say so once with the concrete failure mode, propose the safer alternative, and record the decision in `/docs/DECISIONS.md`.
- Never invent data, prices, sources or model names. If something is unknown, the app must show "unavailable", not a guess.

1. What the app does (one paragraph)

The user picks any tradable "product" (stock, index, ETF, F&O contract, commodity, currency pair, metal or crypto) from Indian and global markets. The system runs a 10-stage analysis. Each stage produces a 1–100 score with reasons and sources. A weighted engine combines them into a 10–1000 composite score with a Buy / Sell / No-trade verdict, a trade plan (entry zone, stop-loss, 3–5 take-profit stages, holding-period badge and complete-by date), and a plain-language explanation of how the score was reached. Confirmed trades are monitored by a backend server at a cadence that matches the chosen timeframe, with emergency alerts when the plan breaks. An "AI Recommendation" mode screens a chosen market and proposes the best candidates. All history is stored and synced between iPhone and iPad through iCloud.

2. Non-negotiable engineering rules

1. **Code computes, AI interprets.** LLMs never read raw candle arrays and never do arithmetic that matters. The feature engine computes indicators, patterns, levels, correlations and statistics; the LLM receives compact feature JSON and explains or weighs it.
2. **Strict JSON or nothing.** Every AI stage returns JSON validated against the schemas in `/shared/schemas`. Invalid output → one retry with the validation error → otherwise the stage is marked `unavailable` (score 50, confidence 0) and the report shows that clearly.
3. **Every claim is traceable.** Each reason cites a source (URL + published time) or a feature ID from the feature pack. Uncited reasons are dropped by the validator.
4. **No hard-coded model names, prices or endpoints.** All AI calls go through one `AIProvider` protocol/interface. Models are listed live from the provider after the user enters a key. Tier→model mapping (Fast / Balanced / Deep) lives in Settings. Model prices come from a remote JSON config that the user can edit.
5. **No order placement in v1.** The app suggests; the user executes in the broker app. (Order placement would trigger SEBI retail-algo rules: static IP, Algo ID, daily 2FA, <10 orders/second.)
6. **Secrets live in the Keychain only** (iOS) and in encrypted server secrets (backend). Never in SwiftData, CloudKit, UserDefaults, logs, analytics or crash reports.
7. **The phone is not the monitoring engine.** iOS cannot run reliable 1-minute background jobs. All monitoring, rules and alerts run on the backend and reach the phone through APNs.
8. **Time is explicit.** Store UTC; display exchange time and IST. Use exchange trading calendars and holidays (NSE/BSE/MCX, NYSE/NASDAQ, 24×7 crypto, 24×5 FX).

9. **Full audit trail.** Every analysis stores: product, data-as-of timestamps, feature-pack hash, prompt template version, provider + model ID per stage, raw AI output, validated output, scores, verdict, cost.
10. **Paper mode is the default.** Real-money mode needs an explicit toggle plus Face ID confirmation.
11. **Show uncertainty everywhere.** Each score shows confidence, data freshness, and the historical hit-rate of that score band (calibration) once enough outcomes exist.
12. **Accessibility is not optional.** Dynamic Type, VoiceOver labels on every chart value, Reduce Transparency, Increase Contrast, colour-blind-safe up/down colours.

3. Tech stack (decided — change only with a written reason)

iOS / iPadOS app

- Swift 6, SwiftUI, minimum deployment target iOS 26 / iPadOS 26 (both devices are personal; this unlocks Liquid Glass and current SwiftData). Build with Xcode 26.x.
- SwiftData with CloudKit private database for user data (sync iPhone ↔ iPad).
- Keychain (`kSecAttrSynchronizable = true`, `kSecAttrAccessibleWhenUnlocked`) for AI API keys so they sync via end-to-end-encrypted iCloud Keychain.
- LocalAuthentication (Face ID / passcode) app lock.
- Charts: Swift Charts for gauges, sparklines, portfolio and score history; TradingView Lightweight Charts (Apache-2.0, bundled locally, attribution shown) inside WKWebView for the full interactive candlestick chart.
- WidgetKit (Home Screen, Lock Screen, StandBy), ActivityKit Live Activities for open intraday positions, App Intents for Siri/Shortcuts.
- Swift Concurrency everywhere; no blocking the main actor.

Backend (required)

- Python 3.12, FastAPI, PostgreSQL, Redis, a job scheduler (APScheduler or Celery beat), pandas/numpy + a technical-analysis library for features.
- Hosted on a small Linux VPS in an Indian region (Mumbai/Bangalore) with a static IP, Docker Compose, Caddy (automatic HTTPS), UFW firewall, SSH keys only, unattended security updates, nightly encrypted Postgres backup.
- APNs via token-based auth (.p8 key) for alerts and Live Activity updates.

4. Repository layout

```
/ios/Score1000.xcodeproj
/ios/Score1000/            App target (Features/, Core/, DesignSystem/, Services/)
/ios/Score1000Widgets/     Widgets + Live Activity
/ios/Score1000Tests/       Unit + snapshot tests
/backend/app/              api/, data/, features/, ai/, scoring/, monitor/, notify/, jobs/
/backend/tests/
/shared/schemas/           stage_output.schema.json, trade_plan.schema.json, feature_pack.schema.json
/shared/prompts/           stage_01_news.md ... stage_10_final.md, verifier.md (versioned)
/shared/config/            model_prices.json, scoring_weights.json, alert_rules.json
/docs/                     SPEC.pdf, DECISIONS.md, RUNBOOK.md, DATA_SOURCES.md
/fastlane/                 Fastfile, Matchfile, Appfile
/.github/workflows/        ios-testflight.yml, backend-ci.yml, monthly-rebuild.yml
```

5. Data layer

Implement a `MarketDataProvider` interface on the backend with adapters, chosen per product by a routing table:

- **India (NSE, BSE, NFO, MCX, CDS):** Upstox API v3 (free with an account; historical candles: 1-min–hour from Jan 2022, daily/weekly/monthly from Jan 2000; live WebSocket). Fyers API as

fallback. Daily login: the app opens the broker OAuth page each trading morning; the token is stored only on the backend.

- **Instrument master:** download Upstox's complete instrument file daily, plus global symbol lists from the global provider and crypto lists; build a searchable `instruments` table (symbol, name, exchange, segment, asset class, lot size, tick size, expiry, ISIN, sector).
- **Global equities, indices, FX (incl. USD/INR), XAUUSD/XAGUSD:** Twelve Data (free Basic: 800 calls/day, 8/min) behind a cache; upgrade path EODHD or Twelve Data Grow.
- **Crypto:** Binance public market data; CoinGecko Demo for metadata.
- **News/events:** NSE official RSS (announcements, board meetings, financial results, circulars), Finnhub or Marketaux news, FRED/BLS release calendars; AI web search for stages that need it.
- **Rules:** respect each provider's rate limits with a token-bucket limiter; cache candles in Postgres (immutable closed bars) and Redis (live); mark every series with `as_of` and `source`; never show scraped/unofficial data as "official".
- Record provider terms in `/docs/DATA_SOURCES.md`. Exchange data is for personal use only; the app must not redistribute it.

6. Feature engine (code, not AI)

For each product build a **feature pack** (JSON, < 8k tokens) per timeframe: 1M, 1W, 1D, 1H, 15m, 5m.

- Trend: SMA/EMA stack (20/50/100/200), slope, ADX, higher-highs/lower-lows structure.
- Momentum: RSI, MACD, rate of change, divergences.
- Volatility: ATR, ATR %, Bollinger width, historical volatility percentile.
- Volume: relative volume, OBV trend, delivery % (India, where available).
- Levels: swing highs/lows, support/resistance clusters, 52-week range, round numbers, VWAP and opening range (intraday).
- Patterns: candlestick and chart patterns with this symbol's own historical hit-rate and sample size (backtested in code).
- History (stage 4): CAGR, max drawdowns with dates, regime segmentation (bull/bear/sideways periods with start/end), top 10 largest monthly moves with dates.
- Cross-asset (stage 8): rolling 60/250-bar correlations and lead-lag with the related products.
- Every feature has a stable ID (e.g. `D.RSI14`, `W.SR.zone2`) so AI reasons can cite it.

7. The 10-stage pipeline and model tiers

#	Stage	Code does first	AI tier	Web search	Refresh/cache
1	Product news	Fetch + dedupe articles (last 7–30 days)	Fast (per-article classify) → code aggregates with time decay	Optional	1–4 h
2	Upcoming events	Results dates, dividends, board meetings, splits, macro calendar	Balanced	Yes	Daily
3	Leadership / governance / country	Promoter holding, pledges, auditor, ratings where available	Balanced	Yes	7–30 days
4	Full monthly history + why	Regimes, drawdowns, biggest moves with dates	**Deep**	Yes	30 days
5	Weekly, last 6 months	Weekly features + patterns with hit-rates	Fast	No	Weekly
6	Daily, full history	Daily features + backtested pattern stats	Fast (escalate to Balanced if signals conflict)	No	Daily
7	Global sector/theme news	Sector mapping; shared per sector	Balanced	Yes	Daily, shared by all symbols in the sector
8	Sister/connected products	Relationship map + rolling correlations	Balanced (map, cached 30 d) + Fast (current read)	Yes for map	Daily
9	Today + 1 week outlook	Intraday + daily features, VWAP, ranges	Balanced	Optional	Each run
10	Final multi-timeframe verdict + trade plan	Feature packs 5m→1M + all stage outputs	**Deep** with extended thinking	Yes (re-check)	Each run
V	Verifier	Re-checks numbers, levels and schema against features	Fast (prefer a different provider)	No	Each run

Other jobs:

- Live monitoring (1m–15m): **no LLM per tick.** Deterministic rules engine on the backend; Fast model only to word an alert summary after a trigger.
- Daily review of each open position: Balanced via the provider's batch API (cheaper) after market close.
- Weekly deep re-score of open positions: Deep.
- AI Recommendation mode: code screens the universe → Fast triage of top ~50 → full pipeline on top 5–10.
- "Explain this score" chat: Balanced.

Default mapping (user can change in Settings after models are listed live):

- Fast = Claude Haiku tier (e.g. `claude-haiku-5-5`) / OpenAI nano–mini tier / Gemini Flash-Lite tier
- Balanced = Claude Sonnet tier / OpenAI mid tier / Gemini Flash tier
- Deep = Claude Opus tier (e.g. `claude-opus-5-5`) / OpenAI top tier / Gemini Pro tier

Use prompt caching for the shared system prompt and feature-pack prefix. Enforce a per-run and per-month spend cap from Settings; show estimated cost before running AI Recommendation mode.

8. Stage output schema (all stages)

```
{
  "stage": 1,
  "product_id": "NSE_EQ|INE002A01018",
  "score": 72,
  "direction": "bullish",
  "confidence": 0.65,
  "horizon_relevance": "short|medium|long",
  "summary": "max 60 words",
  "reasons": [
    {"point": "text", "impact": 2, "evidence": ["D.RSI14", "src:3"]}
  ],
  "sources": [
    {"id": "src:3", "title": "", "url": "", "published_at": "ISO-8601"}
  ],
  "data_as_of": "ISO-8601",
  "warnings": []
}
```

Score meaning: 1 = strongly bearish, 50 = neutral, 100 = strongly bullish. `impact` is −3…+3.

9. Scoring engine (deterministic code)

1. Confidence shrink: `s_eff = 50 + confidence × (score − 50)`.
2. Horizon weights from `/shared/config/scoring_weights.json` (each row sums to 100):

Stage	Scalp/Intraday	Swing	Positional	Long-term	Equal
1 News	8	10	10	8	10
2 Upcoming	4	8	10	10	10
3 Leadership	0	3	8	15	10
4 Monthly history	0	4	10	20	10
5 Weekly 6M	2	10	14	10	10
6 Daily	10	15	12	5	10
7 Global sector	4	8	10	12	10
8 Connected	8	8	8	8	10
9 Today + 1W	24	14	6	2	10
10 Final	40	20	12	10	10

3. Composite `C = 10 × Σ(w_i × s_eff_i) / 100` → range 10–1000. Buy score = C; Sell score = 1010 − C.
4. Verdict bands: ≥ 750 Strong Buy · 620–749 Buy · 381–619 No trade / Wait · 251–380 Sell · ≤ 250 Strong Sell.
5. Vetoes (downgrade to "No trade" with the reason shown): stale data beyond the timeframe limit; results/major event inside the holding window (unless the user opts in); liquidity or spread below thresholds; verifier finds a numeric mismatch; fewer than 7 of 10 stages available.
6. Store everything; nightly job records outcomes (did TP1 hit before SL? return at horizon end) and computes calibration per score band and per stage.

10. Trade plan (stage 10 + code)

- Entry zone, stop-loss (structure- or ATR-based), R = |entry − stop|.
- Take-profit ladder (user picks 3, 4 or 5 stages). Default 5-stage: TP1 = 1R (book 30 %, move stop to breakeven) · TP2 = 2R (25 %, trail stop to TP1) · TP3 = 3R (20 %) · TP4 = 4R (15 %) · TP5 = runner with ATR trailing stop (10 %). Code snaps levels to nearby resistance/support.
- Position size = (capital × risk %) / R (default risk 1 %; max 2 %).
- Holding badge: Scalp · Intraday · Short-term (2–10 trading days) · Swing (2–6 weeks) · Positional (1–6 months) · Long-term (6 months+). Complete-by date = entry + max holding; on expiry the app asks:

exit or re-analyse.

- AI may propose levels; code validates ordering (stop < entry < TP1 < … for longs), minimum R:R ≥ 1.5 at TP2, and tick-size rounding.

11. Monitoring and emergency protocol (backend)

Cadence follows the chosen timeframe:

- 1m/5m: broker WebSocket ticks, rules on every tick/bar close, Live Activity updates.
- 15m/1h: every bar close.
- Daily: intraday price-alert rules live + full review after the close.
- Weekly/monthly: price-alert rules live + daily close check + weekly deep review.

Upcoming-trade watch: when price comes within the proximity band of the AI entry zone (default 0.5 × ATR), send "Entry zone reached" with the current score re-checked.

Emergency triggers: E1 stop/invalidation crossed · E2 gap beyond stop or > 2 ATR adverse · E3 adverse velocity > 1.5 ATR within N bars · E4 news shock (stage-1 rescan drop ≥ 25 or keywords: fraud, raid, SEBI order, default, CEO resignation, trading halt) · E5 thesis broken (composite crosses to the opposite band) · E6 data feed or server heartbeat lost ("Monitoring offline") · E7 complete-by date passed.

Emergency actions: time-sensitive push (repeat every 2 min until acknowledged, max 5), red Live Activity, full-screen in-app takeover with "Close now" checklist, entry in the journal. Always remind the user to keep a real stop-loss/GTT order at the broker — push delivery can be delayed.

12. Data model (SwiftData, CloudKit-compatible)

Rules: no `@Attribute(.unique)`; every property has a default or is optional; every relationship optional; no `.deny` delete rule; schema is additive after production promotion.

Entities: `Product`, `Watchlist`, `AnalysisRun`, `StageResult`, `TradePlan`, `TakeProfitLevel`, `Position` (planned → entered → partially closed → closed), `Fill`, `PositionReview`, `AlertEvent`, `RecommendationRun`, `CalibrationSnapshot`, `UserSettings`. Large raw payloads (feature packs, raw AI output) stay on the backend; the app stores IDs + summaries.

13. UI / UX

- **Liquid Glass (iOS 26):** let system navigation, tab bars, toolbars and sheets adopt glass automatically. Use `.glassEffect()` only for floating controls (score chip, quick-action bar, timeframe picker), group them in `GlassEffectContainer`, morph with `glassEffectID(_:in:)`, buttons with `.buttonStyle(.glass)` / `.glassProminent`. Never put glass behind dense numbers, tables or candles. When Reduce Transparency is on, fall back to solid surfaces.
- **Themes:** Liquid Glass (default) · Calm Light · Calm Dark (#121417, not pure black) · OLED Night · Paper (warm, low blue) · High Contrast. Up/down colours softened; colour-blind option (blue/orange). Numbers in SF Pro with `.monospacedDigit()`.
- **Screens (iPhone tab bar; iPad `NavigationSplitView`):** Lock → Home (today's alerts, open positions, upcoming trades) → Markets (search all products, filters by exchange/asset class/sector, watchlists) → Analyse (live stage-by-stage progress cards, cancellable) → Score Report (0–1000 gauge, verdict, 10 stage cards each with score, reasons, sources, model used; "How this score was calculated" breakdown with weights) → Trade Plan (chart with entry/stop/TP levels, badge, complete-by date, size calculator) → Upcoming Trades → Positions → Journal (invested, realised/unrealised P&L, win rate, average R, AI calibration) → AI Recommendations → Alerts Center → Settings (AI providers & keys, model tiers, spend caps, risk %, themes, lock, data sources, paper/real mode).
- Motion: subtle, 200–300 ms, respects Reduce Motion. Loading states are skeletons shaped like the content; every AI wait has a cancel button.

14. Security checklist

- Face ID/passcode on launch and after 30 s in background; privacy cover in the app switcher; Face ID re-confirm for viewing keys, switching to real-money mode, and exporting data.
- AI keys: Keychain (synced). If server-side jobs need them, send once over TLS to the backend and store encrypted at rest; a Settings switch shows which keys the server holds and can wipe them.
- Backend auth: Sign in with Apple + single-user allowlist + short-lived JWT, or a Secure Enclave device key signing each request. Certificate pinning (`NSPinnedDomains`) for your own backend only. ATS on, no arbitrary loads.
- Prompt-injection defence: news/web text is wrapped as untrusted data; models get no tools that act on accounts; outputs schema-validated; numbers recomputed by code.
- Server hardening as in §3; 2FA on Apple ID, GitHub, broker, VPS and AI consoles; secrets in GitHub Actions encrypted secrets.
- Privacy-sensitive redaction for widgets/Live Activities on the Lock Screen (option to hide P&L).

15. CI/CD: GitHub → TestFlight → iPhone/iPad

- `ios-testflight.yml` on `runs-on: macos-26`; select a pinned Xcode 26.x; fastlane `match` (private certs repo) + App Store Connect API key (.p8, Key ID, Issuer ID as secrets); `build_app` → `upload_to_testflight` to the **internal** group (no Beta App Review).
- Bump build number from the run number. Run unit tests before upload.
- `monthly-rebuild.yml`: scheduled rebuild + upload so a TestFlight build never hits the 90-day expiry.
- `backend-ci.yml`: lint, tests, build Docker image, deploy over SSH to the VPS on tag.

16. Milestones and acceptance tests

Work milestone by milestone. At the start of each: write a short plan. At the end: run tests, update `/docs/DECISIONS.md`, and list what the user must do manually (accounts, keys, approvals).

- **M1 Foundation:** project, design system + 6 themes, Face ID lock, Keychain key storage, AI provider screen that lists live models for Anthropic/OpenAI/Gemini, CI uploading to TestFlight. ✓ App installs on iPhone and iPad from TestFlight; keys sync; lock works.
- **M2 Backend + data:** VPS, auth, Upstox login flow, instrument master, candles + caching, global/crypto adapters, product search, candlestick chart. ✓ Search finds RELIANCE (NSE), XAUUSD, AAPL, BTCUSDT; charts load 1m→1M.
- **M3 Feature engine + backtest harness:** all features, pattern hit-rates, feature-pack JSON. ✓ Golden tests against hand-checked values.
- **M4 AI stages + scoring:** 10 stages + verifier, schemas, weights, vetoes, Score Report UI with explanations and sources. ✓ Invalid AI JSON never reaches UI; each reason cites evidence; cost per run logged.
- **M5 Trade plan + trades:** TP ladder, sizing, badges, Upcoming Trades, Positions, Journal with P&L. ✓ Level ordering validator; P&L matches manual calculation.
- **M6 Monitoring + alerts:** rules engine, APNs, Live Activities, widgets, emergency protocol, heartbeat. ✓ Simulated feed triggers each E1–E7 alert on device within 10 s.
- **M7 AI Recommendation mode:** screener → triage → full pipeline, cost preview. ✓ Runs within the spend cap.
- **M8 Calibration + hardening:** outcome tracking, calibration charts, security review, failure drills (kill data feed, kill AI provider, expire broker token). ✓ App degrades with clear messages, never shows a fabricated score.

Paper-trade for at least 4–8 weeks and review calibration before using real-money mode.

10. Sources

Facts were checked on 4 November 2026. Items marked † come from secondary or dated sources and should be re-checked before relying on them.

AI models & pricing

- Anthropic pricing — platform.claude.com/docs/en/about-claude/pricing
- Anthropic models list API — platform.claude.com/docs/en/api/models/list
- Gemini models API — ai.google.dev/api/models; Gemini pricing — ai.google.dev/gemini-api/docs/pricing †
- OpenAI, xAI, Perplexity model prices — third-party trackers (morphllm.com, costbench.com, Oct 2026) †
- PriceSeer: LLM stock-direction benchmark — arxiv.org/html/2601.06088
- TSAIA: time-series reasoning benchmark — arxiv.org/pdf/2509.01822
- FAITH: financial table calculation errors (ICAIF '25) — arxiv.org/html/2508.05201

Market data & news

- Upstox trading API (free) — upstox.com/trading-api; historical candles v3 — upstox.com/developer/api-documentation/v3/get-historical-candle-data
- Upstox instrument master — assets.upstox.com/market-quote/instruments/exchange/complete.json.gz
- Fyers API — fyers.in/fyers-api
- Zerodha Kite API charges — support.zerodha.com (Kite API charges article)
- Dhan Data API subscription — dhan.co/support (DhanHQ data API) †
- Twelve Data pricing — twelvedata.com/pricing; Finnhub pricing — finnhub.io/pricing; EODHD plans †
- NSE RSS feeds — nseindia.com/static/rss-feed; NSE data policy — nseindia.com/market-data/nse-data-policy
- NSE cease-and-desist to apps — yourstory.com (Oct 2022)
- TradingView Lightweight Charts licence — npmjs.com/package/lightweight-charts

Apple platform

- SwiftData iCloud sync — developer.apple.com/documentation/swiftdata (Syncing model data across a person's devices)
- kSecAttrSynchronizable — developer.apple.com/documentation/security/ksecattrsynchronizable
- Background task timing — developer.apple.com/forums/thread/724506
- Time-sensitive notifications & Critical Alerts entitlement — developer.apple.com/documentation/usernotifications
- Swift Charts candlesticks (RuleMark + RectangleMark) — developer.apple.com/forums/thread/778613
- Upload requirement Xcode 26 / iOS 26 SDK from 28 Apr 2026 — developer.apple.com/news/upcoming-requirements
- GitHub macos-26 runners GA — github.blog/changelog (26 Feb 2026)
- TestFlight limits — developer.apple.com/help/app-store-connect (TestFlight overview)
- App Store Review Guidelines 1.1.6, 2.5.4, 3.2.1(viii), 5.1.1(ix) — developer.apple.com/app-store/review/guidelines
- Apple Developer fee in India (2024 forum post) — developer.apple.com/forums/thread/746357 †
- Apple Stocks app / Liquid Glass references — 9to5mac.com/2020/08/16/stocks-app; medium.com (iOS 26 Liquid Glass in SwiftUI reference)

Regulation

- SEBI circular "Safer participation of retail investors in Algorithmic trading", 4 Feb 2025 — sebi.gov.in
- Implementation timeline and 10 orders/second threshold — algobulls.com; quantinsti.com; finseclaw.com †
- Fyers notice on SEBI framework from 1 Apr 2026 — fyers.in/notice-board
- SEBI RA (Third Amendment) Regulations 2024 — scconline.com (19 Dec 2024) †